

A certified multilevel lower bound for the Galerkin prediction error of a log-singular Toeplitz form with a prime-power comb

Stefan Hamann

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Abstract

For a reflection-odd compression A of a finite Toeplitz form and a data vector b we study the prediction defect $\varepsilon = 1 - b^\top A^{-1}b$, which is the Galerkin error of a piecewise-constant approximation to a fixed function, and we produce *lower* bounds for it on a nested ladder of resolutions. The mechanism is a change of direction: written as a saturation telescope, the quantity to be supplied at each level is a residual measured in the *inverse* energy norm, hence a *maximum* over test vectors, so a single explicit test vector already yields a valid lower bound and the coarse-level block enters only through its sign — never through its smallest eigenvalue, its condition number, or a strengthened Cauchy–Schwarz constant. Combining this certified rung with a margin-free bordering criterion, we obtain a chain that transports semidefiniteness along a nested family of windows, and we exhibit a verified instance in which every load-bearing step is a completed Cholesky factorisation above an explicitly declared floating-point floor: 430 such certificates over 52 windows and 88 rungs. Two features of the result are stated up front rather than in a remark. First, an accounting convention: the numerator of the certified rung is a function of the coarse solution carried along the ladder, so the bound is a bound for the carried-solution rung. Second, a measured negative result: the solution-free variant of the same rung fails, not by a small margin but by cancellation — the coupling term annihilates the explicit high-pass part of the data instead of perturbing it, and a norm-only Cauchy–Schwarz estimate of it returns exactly zero of the rung on every rung of the tested ladder. The example class that motivates the construction has a symbol built from a logarithmic profile minus an explicit cosine comb at the frequencies $\log n$, n a prime power; the arithmetic enters only as the source of the comb positions and only through two classical gap inequalities.

1 Introduction

1.1 The problem, and why the direction matters

Let $A \in \mathbb{R}^{h \times h}$ be symmetric positive definite, let $b \in \mathbb{R}^h$, and set

$$\varepsilon = 1 - b^\top A^{-1}b, \quad Q = A - bb^\top. \quad (1)$$

The scalar ε is the value of the quadratic functional $J(v) = 1 - 2b^\top v + v^\top A v$ at its minimiser $u = A^{-1}b$; when A is a Galerkin matrix on a finite-dimensional space V and b the load vector of a fixed right-hand side, ε is exactly the Galerkin error of the best approximation in V , measured in the energy of the ambient form. When A is a Toeplitz matrix and b the first coordinate vector, ε is the classical one-step prediction error of Szegő, Levinson and Durbin [27, 16, 21, 12], whence the name we use throughout. Finally, and this is the elementary fact the whole paper turns on, by Albert’s criterion applied to the two Schur readings of the bordered matrix $\begin{bmatrix} A & b \\ b^\top & 1 \end{bmatrix}$ one has, for $A \succ 0$,

$$\varepsilon \geq 0 \iff \begin{bmatrix} A & b \\ b^\top & 1 \end{bmatrix} \succeq 0 \iff Q \succeq 0. \quad (2)$$

So a *lower* bound on ε is a statement about the semidefiniteness of a rank-one-perturbed form, and vice versa.

Upper bounds for ε are the classical direction. Céa’s lemma, duality, and the Bramble–Hilbert machinery all estimate an approximation error from above [8]. Lower bounds are the hard direction, and the standard device for obtaining them — the *saturation assumption*, which asserts that refining the space reduces the error by a factor bounded away from one — is known to be delicate: Bank and Smith made it the basis of hierarchical a posteriori estimators [3], Bornemann, Erdmann and Kornhuber isolated the constants it consumes [5], and Dörfler and Nochetto showed that it holds under a smallness condition on the data oscillation and can fail otherwise [10]. In the multilevel literature the same difficulty appears as the strengthened Cauchy–Schwarz constant $\gamma^2 < 1$ of hierarchical splittings [32, 6, 2, 7]: it is a *worst-case* quantity, an upper bound on a coupling, and any bound that has to supply it inherits its worst case.

There is a second, more specific obstruction in the Toeplitz setting, and it is what makes the example class of Section 2.4 genuinely hard rather than merely technical. If the symbol σ of the form is nonnegative, the Toeplitz matrix $T_M(\sigma)$ is positive semidefinite, and conversely a pointwise minorant $\ell \leq \sigma$ transfers to an operator minorant $T_M(\ell) \preceq T_M(\sigma)$; this is Parseval together with the Fejér–Riesz representation of nonnegative trigonometric polynomials [13, 26], and it is the standard route from a symbol to a spectral bound [19, 16]. The route is one-sided in a way that matters here. For the forms we care about the smallest eigenvalue is not produced by a small value of the symbol at some frequency; it is produced by *cancellation inside the quadratic form*, along a direction that is smooth and lives in the coarse part of the space. No pointwise minorant of the symbol can see such a direction, because a pointwise minorant is by construction blind to which vector realises the minimum. Section 2.4 records the measured form of this statement for the example class; the theorem of Section 4 is designed so that it never needs a symbol minorant at all.

1.2 Contribution

- (C1) **A direction reversal that makes a certified rung possible** (Theorem 4.4). On a nested ladder, the increment $\delta_l = \varepsilon_l - \varepsilon_{l+1}$ of the saturation telescope is the squared *inverse-energy* norm of the level residual, hence a maximum over test vectors. A single test vector therefore supplies a valid lower bound, and the denominator of that bound wants the relevant Schur complement from *above*. Schur complement monotonicity [17] supplies the upper bound from the mere semidefiniteness of the coarse block, Parseval supplies an upper bound by a certified majorant of the symbol, and Löwner’s antitony of the matrix inverse [22] closes the chain. No coarse inverse, no $\lambda_{\min}$ of a coarse block, and no strengthened Cauchy–Schwarz constant enters. The optimal test vector for the majorant is available in closed form.
- (C2) **A margin-free composition of the certified rungs into a chain** (Corollary 5.6). Two exact identities (Lemmas 5.2 and 5.3) show that on a nested family of windows at one common resolution the trailing block of the next window’s form *is* the current window’s form. Albert’s criterion [1] with Douglas’ range condition [11] then transports semidefiniteness from one window to the next using only the *sign* of that block — never the size of ε — and the residual condition is a Schur complement of the small dimension of the window increment.
- (C3) **An honest boundary, in both directions.** The certified rung is stated for the carried-solution numerator, and this accounting convention is part of the hypothesis list, not a remark (Convention 4.7). The solution-free variant is then shown, by measurement, to fail through cancellation (Proposition 6.1); we report this as a negative result about a specific route on a specific ladder, since it is what tells a reader that the convention

was not an oversight. Section 8 states what the construction does not give, the principal item being uniformity in the window index.

Everything load-bearing is verified numerically in Section 7: a single script, runtime about ten seconds, produces 430 completed Cholesky certificates and the identity residuals, each stated relative to a declared floating-point floor in the sense of Wilkinson and Higham [31, 18].

1.3 Notation

$T_M(c) = (c_{|r-s|})_{r,s=0}^{M-1}$ is the symmetric Toeplitz matrix of a real sequence $c = (c_m)_{m \geq 0}$. For real integrable even g on $[-\pi, \pi]$ with Fourier coefficients $\hat{g}_m$ we write $T_M(g) := T_M(\hat{g})$; no confusion arises because a symbol enters only through its first M coefficients. $X \succeq 0$ and $X \succ 0$ denote positive semidefiniteness and definiteness, $X \preceq Y$ means $Y - X \succeq 0$, X^+ is the Moore–Penrose pseudoinverse, and $\|v\|_X^2 = v^\top X v$ for $X \succeq 0$. Finally $(t)_+ = \max(t, 0)$ and $\Delta_D(s) = (1 - |s|/D)_+$ is the unit-height hat of half-width D .

2 Setting

2.1 The odd window form and its defect

Fix a cell width $D > 0$ and an even $M \in 2\mathbb{N}$, and put $\alpha = MD/2$ and $h = M/2$. The grid is the partition of $(-\alpha, \alpha)$ into M cells of width D with midpoints

$$\bar{x}_r = -\alpha + (r + \tfrac{1}{2})D, \quad r = 0, \dots, M-1, \quad (3)$$

and χ_r denotes the L^2 -normalised indicator of cell r , i.e. $\chi_r = D^{-1/2} \mathbf{1}_{[\bar{x}_r - D/2, \bar{x}_r + D/2]}$. Let $J \in \mathbb{R}^{M \times M}$ be the reflection $(Jv)_r = v_{M-1-r}$ and let

$$B \in \mathbb{R}^{M \times h}, \quad B e_r = \frac{1}{\sqrt{2}} (e_r - e_{M-1-r}), \quad r = 0, \dots, h-1, \quad (4)$$

be the isometry onto the reflection-odd sector, so $B^\top B = I_h$ and $JB = -B$.

Definition 2.1 (window form and pole vector). Given a real sequence $c = (c_m)_{m=0}^{M-1}$, the *odd window form* and the *pole vector* at (D, M) are

$$A = B^\top T_M(c) B \in \mathbb{R}^{h \times h}, \quad A_{rs} = c_{|r-s|} - c_{M-1-r-s}, \quad (5)$$

$$b = (b_r)_{r=0}^{h-1}, \quad b_r = \frac{8}{\sqrt{D}} \sinh\left(\frac{D}{4}\right) \sinh\left(\frac{\bar{x}_r}{2}\right). \quad (6)$$

The associated *defect* and *bordered form* are ε and Q of (1).

The particular constant in (6) is not a normalisation choice; it is the reason the whole construction is a Galerkin problem on nested spaces.

Lemma 2.2 (b is the coefficient vector of a fixed function). Let $f(x) = 2 \sinh(x/2)$. Then $b_r = \langle f, \chi_r \rangle_{L^2(\mathbb{R})}$ for $r = 0, \dots, h-1$, for every $D > 0$ and every M .

Proof. $\langle f, \chi_r \rangle = D^{-1/2} \int_{\bar{x}_r - D/2}^{\bar{x}_r + D/2} 2 \sinh(x/2) dx = D^{-1/2} [4 \cosh(x/2)]_{\bar{x}_r - D/2}^{\bar{x}_r + D/2}$, and $\cosh(a+d) - \cosh(a-d) = 2 \sinh a \sinh d$ with $a = \bar{x}_r/2$, $d = D/4$ gives $8D^{-1/2} \sinh(D/4) \sinh(\bar{x}_r/2)$. $\square$

Remark 2.3. Write $V_{D,M} \subset L^2(-\alpha, \alpha)$ for the space of odd piecewise-constant functions on the grid (3); $\{B\chi\}$ is an orthonormal basis of it, and $\dim V_{D,M} = h$. If $a(\cdot, \cdot)$ denotes the bilinear form whose matrix in that basis is A , then by Lemma 2.2

$$\varepsilon = \min_{v \in V_{D,M}} \left(1 - 2\langle f, v \rangle + a(v, v) \right), \quad (7)$$

so ε is the Galerkin defect of the fixed function f in the space $V_{D,M}$, and its minimiser is the Galerkin solution $u = A^{-1}b$. Nothing in (7) refers to c except through a .

2.2 The ladder and exact nesting

We refine at fixed α . Let $L \geq 1$ and

$$D_l = 2^{-l} D_0, \quad M_l = 2^l M_0, \quad h_l = M_l/2, \quad l = 0, \dots, L, \quad (8)$$

so $\alpha = M_l D_l/2$ is the same at every level and $V_{D_0, M_0} \subset V_{D_1, M_1} \subset \dots \subset V_{D_L, M_L}$. Write $A^{(l)}, b^{(l)}, \varepsilon_l, u^{(l)}$ for the objects of Definition 2.1 at level l . Let $P_l : \mathbb{R}^{h_l} \rightarrow \mathbb{R}^{h_{l+1}}$ be the matrix of the inclusion $V_{D_l, M_l} \hookrightarrow V_{D_{l+1}, M_{l+1}}$ in the orthonormal bases, i.e. the L^2 -normalised piecewise-constant prolongation: each coarse cell splits into two fine cells and

$$(P_l x)_{2i} = (P_l x)_{2i+1} = \frac{1}{\sqrt{2}} x_i. \quad (9)$$

We suppress the level index on P when it is clear. Let $Z_l : \mathbb{R}^{h_l} \rightarrow \mathbb{R}^{h_{l+1}}$ be the complementary Haar injection,

$$(Z_l x)_{2i} = \frac{1}{\sqrt{2}} x_i, \quad (Z_l x)_{2i+1} = -\frac{1}{\sqrt{2}} x_i, \quad (10)$$

so that

$$P^\top P = Z^\top Z = I_{h_l}, \quad P^\top Z = 0, \quad \text{ran } P \oplus \text{ran } Z = \mathbb{R}^{h_{l+1}}. \quad (11)$$

The pair (P, Z) is a two-level orthogonal splitting of the fine space into its piecewise-constant (coarse) and oscillatory (high-pass) parts.

Assumption 2.4 (exact nesting). For every $l < L$,

$$P^\top A^{(l+1)} P = A^{(l)}, \quad P^\top b^{(l+1)} = b^{(l)}. \quad (12)$$

Assumption 2.4 is not restrictive; it is exactly the statement that the ladder is one bilinear form on nested spaces. Its second half is unconditional.

Lemma 2.5 (nesting of the pole vector). $P^\top b^{(l+1)} = b^{(l)}$ holds for every D_l and every M_l .

Proof. Immediate from Lemma 2.2, since P is the matrix of an inclusion of spaces in orthonormal bases and f is fixed. Directly: with $d = D_l/2$ the two fine midpoints inside the coarse cell centred at $\bar{x}$ are $\bar{x} \mp d/2$, so

$$(P^\top b^{(l+1)})_i = \frac{1}{\sqrt{2}} \cdot \frac{8}{\sqrt{d}} \sinh \frac{d}{4} \left[\sinh \frac{\bar{x} - d/2}{2} + \sinh \frac{\bar{x} + d/2}{2} \right] = \frac{1}{\sqrt{2}} \cdot \frac{8}{\sqrt{d}} \sinh \frac{d}{4} \cdot 2 \cosh \frac{d}{4} \sinh \frac{\bar{x}}{2},$$

which equals $8(2d)^{-1/2} \sinh(d/2) \sinh(\bar{x}/2) = b_i^{(l)}$ because $D_l = 2d$. $\square$

The first half of (12) is a property of the sequence c : it holds whenever c_m at level l is the exact lag sequence of a fixed kernel against the cell basis at width D_l , so that the coarse form is the fine form restricted to the coarse subspace. For the example class of Section 2.4 this is a closed-form identity in D ; in the verification of Section 7 it is checked to relative accuracy $3.0 \cdot 10^{-14}$ on every rung and reported as such.

2.3 The two-level split of the fine form

Fix a level l and abbreviate $A_f = A^{(l+1)}$, $V = P$, $W = Z$. Setting

$$A_c = V^\top A_f V, \quad A_z = W^\top A_f W, \quad B_x = V^\top A_f W, \quad (13)$$

the orthogonal splitting (11) puts the fine form in block shape $\begin{bmatrix} A_c & B_x \\ B_x^\top & A_z \end{bmatrix}$, and Assumption 2.4 identifies $A_c = A^{(l)}$: *the coarse block of a rung is the window form of the level below it*. The Schur complement onto the oscillatory space is

$$S = A_z - B_x^\top A_c^{-1} B_x. \quad (14)$$

If $A_f \succ 0$ then $A_c = V^\top A_f V \succ 0$ for the isometry V , so (14) is well defined, and $S \succ 0$.

2.4 The example class

The construction above is symbol-agnostic; Theorem 4.4 and Corollary 5.6 use no property of c beyond what is stated in their hypotheses. The class we verify on, and which motivates the whole question, is the following.

Definition 2.6 (the comb class). Let $a : \mathbb{R} \rightarrow \mathbb{R}$ be the even kernel

$$a(s) = -(\gamma + \log \pi) \Delta_D(s) - \Delta_D(s) \log(1 - e^{-2W}) + 2 \int_0^W \frac{\Delta_D(s) e^{-2w} - S_D(s, w) e^{-w/2}}{1 - e^{-2w}} dw, \quad (15)$$

where $W = |s| + D$, $S_D(s, w) = \frac{1}{2}[\Delta_D(s - w) + \Delta_D(s + w)]$, and γ is Euler's constant. Let $\mathcal{U} = \{(u_j, \mu_j)\}_j$ be a finite set of *comb nodes* $u_j > 0$ with weights $\mu_j > 0$, and put

$$c_m = a(mD) - \sum_{j: u_j \leq 2\alpha} \mu_j S_D(mD, u_j), \quad m = 0, \dots, M-1. \quad (16)$$

The *prime-power instance* is $u_j = \log n_j$ over the prime powers $n_j = p^\nu$ in increasing order, with $\mu_j = 2\Lambda(n_j)/\sqrt{n_j}$ and Λ the von Mangoldt weight $\Lambda(p^\nu) = \log p$.

Three comments, all of them limitations of scope rather than claims.

(i) *The label “log-singular”.* The integrand of (15) carries the factor $(1 - e^{-2w})^{-1} = \sum_{n \geq 0} e^{-2nw}$, whose expansion is the digamma series, and the constant is $-\gamma - \log \pi$. The kernel is therefore a logarithmic (digamma) profile smoothed against the hat Δ_D , and the label in the title refers to that profile. The integrand vanishes at $w = 0$ because $S_D(s, 0) = \Delta_D(s)$, so a is finite for every s ; the symbol $\sigma^{(M)}(\theta) = \sum_{|m| \leq M-1} c_{|m|} e^{im\theta}$ is a trigonometric polynomial for each fixed M , and no asymptotic property of the family is used anywhere below. The classical regularity theory for symbols of this type is Widom's and Fisher–Hartwig's [30, 14, 4]; we name it as the address of the questions we do *not* answer, not as an ingredient.

(ii) *What the comb does.* By (16) the symbol of c is a logarithmic profile minus an explicit cosine comb: each node contributes a term proportional to $\cos(u_j \theta / D)$ modulated by the hat's Fourier transform. This is the structural reason a pointwise symbol minorant is useless here. The comb dips are numerous and narrow — narrow enough that the large-sieve budget one would naturally spend on them [23] is exhausted before the window is covered — the infimum of the symbol over a window is driven to zero by them, and the near-null direction of the resulting form is nevertheless *smooth*: it lives in the coarse part $\text{ran } V$ and its Rayleigh quotient is small because of cancellation inside the form, not because the symbol is small along it. The measured statement is in Section 7.

(iii) *What the arithmetic is used for.* Only the positions $u_j = \log n_j$, and only through two classical inequalities on the consecutive log-gaps $g_k = u_{k+1} - u_k$, namely $g_k \leq \log 2$ (Bertrand's postulate in Chebyshev's form [9]) and $g_k \geq \log(1 + 1/n_k)$. Both are used solely to check that a resolution is admissible in the sense of Definition 5.1; no unproved property of the sequence (u_j) enters the construction, and no arithmetic statement is claimed.

3 The saturation telescope

The identities of this section are classical; we record them in the form we use.

Lemma 3.1 (saturation identity and telescope). *Assume $A^{(l)} \succ 0$ for all l and Assumption 2.4. Put*

$$\delta_l = \|u^{(l+1)} - Pu^{(l)}\|_{A^{(l+1)}}^2 \geq 0. \quad (17)$$

Then for every $l < L$

$$\varepsilon_l = \varepsilon_{l+1} + \delta_l, \quad \text{hence} \quad \varepsilon_0 = \varepsilon_L + \sum_{l=0}^{L-1} \delta_l \quad (\text{exactly}). \quad (18)$$

Moreover, with the level residual $r_l = b^{(l+1)} - A^{(l+1)}Pu^{(l)}$,

$$V^\top r_l = 0, \quad \delta_l = \|r_l\|_{(A^{(l+1)})^{-1}}^2. \quad (19)$$

Proof. Let $J_{l+1}(v) = 1 - 2(b^{(l+1)})^\top v + v^\top A^{(l+1)}v$ on $\mathbb{R}^{h_{l+1}}$. Completing the square at its minimiser $u^{(l+1)}$ gives $J_{l+1}(v) = \varepsilon_{l+1} + \|v - u^{(l+1)}\|_{A^{(l+1)}}^2$ for all v . Take $v = Pu^{(l)}$. By Assumption 2.4, $J_{l+1}(Pu^{(l)}) = 1 - 2(b^{(l)})^\top u^{(l)} + (u^{(l)})^\top A^{(l)}u^{(l)} = \varepsilon_l$, which is (18); the telescope follows by summation. For (19), Assumption 2.4 gives $V^\top r_l = b^{(l)} - A^{(l)}u^{(l)} = 0$, which is Galerkin orthogonality, and $A^{(l+1)}(u^{(l+1)} - Pu^{(l)}) = r_l$ turns (17) into (19). $\square$

Two consequences of (19) deserve to be separated, because the whole difficulty of the subject sits in the difference between them.

Lemma 3.2 (the two dual descriptions of a rung). *In the notation of Section 2.3, put*

$$s_l = W^\top b^{(l+1)}, \quad \hat{s}_l = s_l - B_x^\top u^{(l)} = W^\top r_l. \quad (20)$$

Then

$$\delta_l = \min_{x \in \mathbb{R}^{h_l}} (Vx + Wy^\star)^\top A^{(l+1)}(Vx + Wy^\star) \quad \text{for a suitable } y^\star, \quad (21)$$

a minimum, and

$$\delta_l = \hat{s}_l^\top S^{-1} \hat{s}_l = \max_{v \neq 0} \frac{(\hat{s}_l^\top v)^2}{v^\top S v}, \quad (22)$$

a maximum.

Proof. $\delta_l = \|u^{(l+1)} - Pu^{(l)}\|_{A^{(l+1)}}^2$ is the squared energy distance from $u^{(l+1)}$ to the coarse space, which is (21). For (22), (19) gives $r_l \in \text{ran } W$, so $r_l = W\hat{s}_l$ by (11), and $\delta_l = r_l^\top (A^{(l+1)})^{-1} r_l = \hat{s}_l^\top (W^\top (A^{(l+1)})^{-1} W) \hat{s}_l$. The block inverse formula for the splitting (13) gives $W^\top (A^{(l+1)})^{-1} W = S^{-1}$. The variational identity $\hat{s}^\top S^{-1} \hat{s} = \max_v (\hat{s}^\top v)^2 / (v^\top S v)$ for $S \succ 0$ is Cauchy–Schwarz in the S -inner product, with equality at $v = S^{-1} \hat{s}$. $\square$

Remark 3.3 (the direction, stated pedantically). From (21), every test vector yields an *upper* bound on δ_l ; a minimum principle cannot supply a lower bound, no matter how good the test vector. From (22), every test vector yields a *lower* bound on δ_l , provided the denominator $v^\top S v$ is replaced by something *larger*. This is the direction reversal of the title of Section 4: what a lower bound on δ_l needs is an *upper* bound on S , and upper bounds on Schur complements are cheap.

4 The certified rung

We need two classical upper-bound mechanisms.

Lemma 4.1 (Schur complement monotonicity). *With the notation of (13) and $A_c \succ 0$, one has $S \preceq A_z$, and the only property of A_c used is $A_c^{-1} \succeq 0$, i.e. the sign of A_c .*

Proof. $A_z - S = B_x^\top A_c^{-1} B_x \succeq 0$ whenever $A_c^{-1} \succeq 0$. See [17]. $\square$

Lemma 4.2 (Parseval majorant). *Let g_1, g_2 be real, even and integrable on $[-\pi, \pi]$ with $g_1 \leq g_2$ almost everywhere. Then $T_M(g_1) \preceq T_M(g_2)$, and consequently $R^\top T_M(g_1) R \preceq R^\top T_M(g_2) R$ for every real matrix R with M rows.*

Proof. For $v \in \mathbb{R}^M$ and $\mathcal{V}(\theta) = \sum_r v_r e^{ir\theta}$, Parseval gives $v^\top T_M(g)v = \frac{1}{2\pi} \int_{-\pi}^{\pi} g(\theta) |\mathcal{V}(\theta)|^2 d\theta$. Apply this to $g = g_2 - g_1 \geq 0$. (That every nonnegative trigonometric polynomial is such a $|\mathcal{V}|^2$ is the Fejér–Riesz theorem [13, 26]; only the displayed identity is needed here.) $\square$

Definition 4.3 (certified majorant). Let $\sigma^{(M)}(\theta) = \sum_{|m| \leq M-1} c_{|m|} e^{im\theta}$, so that $T_M(c) = T_M(\sigma^{(M)})$. A real even $\text{up} \in L^1[-\pi, \pi]$ is a *certified majorant* at level $l+1$ if $\text{up} \geq \sigma^{(M_{l+1})}$ pointwise. We then set

$$U_l = W^\top B^\top T_{M_{l+1}}(\text{up}) B W. \quad (23)$$

Note that $A_z = W^\top B^\top T_{M_{l+1}}(c) B W$ by (5) and (13), so Lemma 4.2 applied with $R = BW$ gives $A_z \preceq U_l$ directly, and the Hankel-type term in (5) never has to be estimated: it is the reflection half of the isometry B .

Theorem 4.4 (the certified rung). *Fix a level l and assume*

- (H1) $A^{(l+1)} \succ 0$;
- (H2) *Assumption 2.4 at level l , so that $A_c = A^{(l)}$;*
- (H3) *up is a certified majorant at level $l+1$ and $U_l \succ 0$.*

Then, with $\hat{s}_l$ of (20),

$$\delta_l \geq \hat{s}_l^\top U_l^{-1} \hat{s}_l =: \hat{c}_l \geq 0, \quad (24)$$

with equality in the second place only if $\hat{s}_l = 0$. The bound is attained by the explicit test vector $v^\star = U_l^{-1} \hat{s}_l$ in (22) after replacing S by U_l . The proof uses no inverse, no smallest eigenvalue and no condition number of A_c , and no strengthened Cauchy–Schwarz constant.

Proof. By (H1) and Section 2.3, $A_c \succ 0$ and $S \succ 0$, so Lemma 3.2 applies and $\delta_l = \hat{s}_l^\top S^{-1} \hat{s}_l$. By Lemma 4.1, $S \preceq A_z$; here (H2) is what makes $A_c \succeq 0$ the conclusion of the level below rather than a new hypothesis. By Definition 4.3 and Lemma 4.2, $A_z \preceq U_l$. Hence $0 \prec S \preceq U_l$, and Löwner’s antitony of the matrix inverse on the positive definite cone [22] gives $U_l^{-1} \preceq S^{-1}$, i.e.

$$\delta_l = \hat{s}_l^\top S^{-1} \hat{s}_l \geq \hat{s}_l^\top U_l^{-1} \hat{s}_l = \hat{c}_l.$$

Positivity of $\hat{c}_l$ for $\hat{s}_l \neq 0$ is $U_l \succ 0$. For the test vector, substituting $v = U_l^{-1} \hat{s}_l$ into $(\hat{s}_l^\top v)^2 / (v^\top U_l v)$ returns $\hat{s}_l^\top U_l^{-1} \hat{s}_l$ exactly, so the closed-form vector is optimal for the majorant. $\square$

Remark 4.5 (what is certifiable, and how). Each of (H1)–(H3) reduces to a completed Cholesky factorisation: $A^{(l+1)} \succ 0$ and $U_l \succ 0$ directly, and the majorant property through $T_{M_{l+1}}(\text{up}) - T_{M_{l+1}}(c) \succeq 0$, which is Lemma 4.2 in the certified direction. In particular $A_c \succeq 0$ — the hypothesis whose quantitative version blocks the classical routes — is supplied by the previous rung’s own conclusion and never has to be quantified. This is what makes a coarse-to-fine induction possible; see Corollary 5.6.

Remark 4.6 (the price of one test vector). The bound (24) is the exact value of the maximum (22) with S replaced by the majorant U_l , so the entire loss relative to δ_l is the loss in $S \preceq U_l$ and none of it is a loss in the choice of test vector. The classical estimate of what a single test vector costs against the optimal one [20] is therefore quoted here only for the contrast: it does not enter.

Convention 4.7 (carried-solution accounting). The numerator $\hat{s}_l = s_l - B_x^\top u^{(l)}$ of (24) is a function of the coarse Galerkin solution $u^{(l)}$ carried along the ladder. Theorem 4.4 and Corollary 5.6 are therefore statements about the *carried-solution* rung: given the ladder and its coarse solutions, the bound is certified. A *solution-free* rung — one whose right-hand side is computable from (c, b, up) alone — is not claimed. See Proposition 6.1.

5 The composed chain

Theorem 4.4 bounds one rung. Two further ingredients turn the rungs into a statement that travels: the telescope of Lemma 3.1, which needs only semidefiniteness at the finest level, and a bordering step that grows the window.

5.1 Windows at one common resolution

Definition 5.1 (admissible common resolution). Let $u_i < u_{i+1} < \dots < u_j$ be consecutive comb nodes with log-gaps $g_k = u_{k+1} - u_k$, and fix $\nu \geq 4$. Put

$$D = \frac{\min_k g_k}{2\nu}, \quad M_k = \min\{M \in 2\mathbb{N} : MD > u_k + D\}, \quad \alpha_k = \frac{M_k D}{2}, \quad (25)$$

and let $A^{(k)}, b^{(k)}, \varepsilon_k, Q^{(k)}$ be the objects of Definition 2.1 at (D, M_k) with the lag sequence (16). We call D *admissible* for the run and write $g_c^{(k)} = (M_{k+1} - M_k)/2$ for the window increment per end.

The resolution is common to the whole run *by construction*, and this is the point: distinct nodes carry distinct natural resolutions $g_k/(2\nu)$, and those do not refine onto one another (a ratio $g_k/g_{k'}$ is a dyadic power only exceptionally — on the verified instance no consecutive gap ratio is dyadic, see Section 7). Choosing one D for the whole run buys an exactly nested family of windows at the price of the run length, since D is fixed by the *smallest* gap it contains.

Lemma 5.2 (the incoming node contributes exactly nothing). *Let D be admissible and let u be a node with $u > 2\alpha_k = M_k D$ (i.e. a node not admitted at window k). Then*

$$S_D(mD, u) = 0 \quad \text{for all } 0 \leq m \leq M_k - 1. \quad (26)$$

Consequently $c_m^{(k+1)} = c_m^{(k)}$ for all $0 \leq m \leq M_k - 1$.

Proof. $S_D(s, u) = \frac{1}{2}[\Delta_D(s - u) + \Delta_D(s + u)]$ is supported in $\{|s - u| < D\} \cup \{|s + u| < D\}$. For $s = mD \geq 0$ and $u > 0$ we have $s + u \geq u \geq \log 2 > D$ on the run (indeed $u \geq u_i$ and $D \leq \log 2/(2\nu)$), so the second hat vanishes. For the first, $m \leq M_k - 1$ gives $s \leq (M_k - 1)D$, hence $u - s \geq M_k D - (M_k - 1)D = D$ and $\Delta_D(s - u) = 0$. The consequence is immediate from (16), because the two windows admit the same nodes with $u_j \leq 2\alpha_k$, all newly admitted nodes satisfy $u > M_k D$, and $a(mD)$ does not depend on M . $\square$

Lemma 5.3 (exact handover identity). *Let $D > 0$, let $M_o < M_n$ be even with $M_n - M_o = 2g_c$, and let $c^{(o)}, c^{(n)}$ be lag sequences with $c_m^{(n)} = c_m^{(o)}$ for $0 \leq m \leq M_o - 1$. Let $A^{(o)}, b^{(o)}$ and $A^{(n)}, b^{(n)}$ be as in Definition 2.1 at (D, M_o) and (D, M_n) , and write $h_o = M_o/2$. Then, with indices $r = g_c + r'$, $s = g_c + s'$ for $r', s' \in \{0, \dots, h_o - 1\}$,*

$$A_{rs}^{(n)} = A_{r's'}^{(o)}, \quad b_r^{(n)} = b_{r'}^{(o)}. \quad (27)$$

Hence the trailing $h_o \times h_o$ block of $Q^{(n)} = A^{(n)} - b^{(n)}(b^{(n)})^\top$ equals $Q^{(o)}$ exactly.

Proof. For the form, $A_{rs}^{(n)} = c_{|r-s|}^{(n)} - c_{M_n-1-r-s}^{(n)}$ by (5). Now $|r - s| = |r' - s'| \leq h_o - 1 \leq M_o - 1$ and $M_n - 1 - r - s = M_n - 2g_c - 1 - r' - s' = M_o - 1 - r' - s'$, which lies in $\{1, \dots, M_o - 1\}$ because $r' + s' \leq 2h_o - 2 = M_o - 2$. Both indices are therefore $\leq M_o - 1$, so the hypothesis on the lag sequences applies and the right-hand side is $c_{|r'-s'|}^{(o)} - c_{M_o-1-r'-s'}^{(o)} = A_{r's'}^{(o)}$. For the pole vector, $\alpha_n - g_c D = (\frac{M_n}{2} - g_c)D = \frac{M_o}{2}D = \alpha_o$, so by (3) $\bar{x}_r^{(n)} = -\alpha_n + (g_c + r' + \frac{1}{2})D = -\alpha_o + (r' + \frac{1}{2})D = \bar{x}_{r'}^{(o)}$, and the prefactor $8D^{-1/2} \sinh(D/4)$ in (6) is unchanged because D is. The statement about Q follows. $\square$

Remark 5.4. The two lemmas together say that on an admissible run the chain *composes literally*: by Lemma 5.2 the lag sequences agree on the old range, so by Lemma 5.3 the trailing block of the next window's bordered form is the current window's bordered form, not merely close to it. The coordinate ordering matters and is the natural one: index $r = 0$ is the outermost cell, so a window that grows by g_c cells per end acquires its new coordinates at the *leading* indices.

5.2 The margin-free bordering step

Lemma 5.5 (Albert’s criterion). *Let $\mathcal{M} = \begin{bmatrix} A & C \\ C^\top & X \end{bmatrix}$ be symmetric. Then $\mathcal{M} \succeq 0$ if and only if*

$$X \succeq 0, \quad \text{ran}(C^\top) \subseteq \text{ran}(X), \quad A - CX^+C^\top \succeq 0. \quad (28)$$

If $X \succ 0$ the middle condition is vacuous and $X^+ = X^{-1}$.

Proof. [1]; the range condition is Douglas’ lemma [11]. $\square$

The point of (28) for us is negative: *nothing in it refers to the size of $\lambda_{\min}(X)$* . Contrast the norm-perturbation surrogate, which writes $\mathcal{M}$ as a perturbation of $\text{diag}(A, X)$ and asks $\lambda_{\min}(X) > \|C\|_2^2 / \lambda_{\min}(A)$ or the symmetric variant; that estimate divides by $\lambda_{\min}(X)$ and is therefore hostage to exactly the quantity a lower bound on ε is trying to produce. Section 7 reports that on the verified instance the surrogate is negative on every step while the exact Schur complement of (28) is of order 10^{-1} .

5.3 The chain

Corollary 5.6 (margin-free chain composition). *Let D be admissible for the consecutive nodes $u_i, \dots, u_j$ in the sense of Definition 5.1. For each k let $D = D_0 > \dots > D_{L_k}$ be the level ladder (8) at fixed α_k , and assume*

- (A1) (base) $\varepsilon_{L_k}^{(k)} \geq 0$ *at the finest level of the chain of window k — equivalently, by (2), $Q^{(k, L_k)} \succeq 0$. This is pure semidefiniteness: no rate, no margin.*
- (A2) *the hypotheses (H1)–(H3) of Theorem 4.4 at every rung of every chain, together with Convention 4.7;*
- (A3) *for every k , the $g_c^{(k)} \times g_c^{(k)}$ Schur complement $A_{\text{new}}^{(k+1)} - C_k (Q^{(k)})^{-1} C_k^\top$ is positive semidefinite, where $A_{\text{new}}^{(k+1)}$ and C_k are the leading block and the coupling block of $Q^{(k+1)}$ in the splitting of Lemma 5.3.*

Then for every $k \in \{i, \dots, j\}$

$$\varepsilon_0^{(k)} \geq \varepsilon_{L_k}^{(k)} + \sum_{l < L_k} \hat{c}_l^{(k)} \geq \sum_{l < L_k} \hat{c}_l^{(k)} > 0, \quad \text{hence} \quad Q^{(k)} \succ 0, \quad (29)$$

and the run is traversed

$$Q^{(i)} \succ 0 \implies Q^{(i+1)} \succ 0 \implies \dots \implies Q^{(j)} \succ 0. \quad (30)$$

Proof. Fix k . By (A2) and Theorem 4.4, $\delta_l^{(k)} \geq \hat{c}_l^{(k)} > 0$ for every rung; by Lemma 3.1, $\varepsilon_0^{(k)} = \varepsilon_{L_k}^{(k)} + \sum_l \delta_l^{(k)}$; and by (A1), $\varepsilon_{L_k}^{(k)} \geq 0$. This is (29), and $Q^{(k)} \succ 0$ follows from (2) with strict inequality. For (30), Lemmas 5.2 and 5.3 put $Q^{(k+1)}$ in the block shape $\begin{bmatrix} A_{\text{new}}^{(k+1)} & C_k \\ C_k^\top & Q^{(k)} \end{bmatrix}$ with the trailing block *equal* to $Q^{(k)}$. Apply Lemma 5.5 with $X = Q^{(k)}$: the first condition is the previous step’s conclusion, the second is vacuous because $Q^{(k)} \succ 0$, and the third is (A3). $\square$

Remark 5.7 (where the induction runs, and why it must). The telescope (18) places its certified bound on the *coarse* level: $\varepsilon_{\text{coarse}} = \varepsilon_{\text{fine}} + \delta$ with $\delta \geq 0$, so the target window must be the coarsest level of its chain and the base case (A1) sits at the finest. A bound at the fine level would require certified *upper* bounds on the rungs, i.e. a strengthened Cauchy–Schwarz constant, which is precisely the quantity the construction is built to avoid. This is a structural feature, not a choice.

6 A measured obstruction: the solution-free rung

Convention 4.7 is the one hypothesis of Corollary 5.6 that is neither an identity nor a Cholesky certificate, so it deserves to be examined rather than declared. The natural attempt to remove it is to replace the optimal test vector $v^\star = U^{-1}\hat{s}$, which needs $u^{(l)}$, by the fully explicit $w = U^{-1}s$, which does not.

Proposition 6.1 (the solution-free rung, and how it fails). *With $w = U_l^{-1}s_l$, the test vector w in (22) gives the valid lower bound*

$$\delta_l \geq \hat{c}_l^{\text{expl}} = \frac{(\hat{s}_l^\top w)^2}{s_l^\top w} = (1 - \rho_l)^2 (s_l^\top U_l^{-1}s_l), \quad \rho_l = \frac{(u^{(l)})^\top B_x w}{s_l^\top U_l^{-1}s_l}, \quad (31)$$

in which the factor $s_l^\top U_l^{-1}s_l$ is computable from (c, b, up) alone and the coarse solution enters through the single scalar ρ_l . Consequently the rung is solution-free up to a constant if and only if ρ_l admits a bound away from 1.

On the verified instance of Section 7 it does not, and the failure is by cancellation rather than by a small margin:

- (i) $|\rho_l|$ is measured in $[0.3888, 14.8018]$ over the 88 rungs of the assembled ladder ($[0.88, 2.61]$ on the separate level chain of the same family), so ρ_l is not a perturbation of 0;
- (ii) $\|\hat{s}_l\|^2/\|s_l\|^2$ is measured in $[6.2 \cdot 10^{-1}, 1.8 \cdot 10^5]$ — below 1 at small α and above it at large α — so the coupling term does not act as a fraction of the explicit high-pass part at all;
- (iii) the norm-only variant, which replaces $\hat{s}_l$ by $\|s_l\| - \|B_x^\top u^{(l)}\|$ under Cauchy–Schwarz, returns exactly 0.0 of δ_l on every one of the 88 rungs.

Proof of the algebraic part. w is a test vector in (22) after replacing S by U_l , which is legitimate by the chain $0 \prec S \preceq U_l$ established in the proof of Theorem 4.4; hence $\delta_l \geq (\hat{s}_l^\top w)^2/(w^\top U_l w)$ and $w^\top U_l w = s_l^\top U_l^{-1}s_l$. Substituting $\hat{s}_l = s_l - B_x^\top u^{(l)}$ gives $\hat{s}_l^\top w = s_l^\top U_l^{-1}s_l - (u^{(l)})^\top B_x w = (1 - \rho_l) s_l^\top U_l^{-1}s_l$, and (31) follows. $\square$

The items (i)–(iii) are measurements on a finite ladder and are reported as such; they do not prove that no solution-free rung exists. What they do establish is the *shape* of the obstruction, and that shape is informative: since ρ_l is of order one and larger, the coupling term $B_x^\top u^{(l)}$ cancels the explicit high-pass part s_l rather than perturbing it. A solution-free rung therefore needs a bound on the *direction* of $\hat{s}_l$, not on its size, and any estimate that proceeds through norms is provably worthless here — item (iii) is that statement, measured.

7 Numerical verification

7.1 Floating-point convention

Every positivity statement below is a completed Cholesky factorisation relative to a declared floor. With $u = 2^{-53} = 1.110 \cdot 10^{-16}$ the unit roundoff and $c_h = (h + 1)/(1 - (h + 1)u)$, a completed Cholesky factorisation of $A - \delta I$ with $\delta = c_h u \|A\|_\infty$ certifies $\lambda_{\min}(A) > 0$ [31, 18]; the cheap certified bound $\|A\|_2 \leq \|A\|_\infty = \max_r \sum_s |A_{rs}|$ is used for the norm [15]. At the largest size occurring below, $h = 1495$, the floor is $1.66 \cdot 10^{-13} \|A\|_\infty$. Quantities obtained from a symmetric eigenvalue routine are labelled *measured* and are never used as certificates; every reported *head-room* is a factor over the declared floor, so any value above 1 certifies the corresponding sign.

quantity	value
common resolution	$D = 3.472446 \cdot 10^{-3}$, from $\min_k g_k = 0.027780$ and $\nu = 4$; effective $\nu_{\text{eff}} = g_k/(2D) = 4.0 \dots 58.4$
run of nodes	30 consecutive prime-power nodes, $n = 2, \dots, 79$; $\alpha_k = 0.3507 \dots 2.1494$; $h_k = 101 \dots 619$; $g_c^{(k)} = 4 \dots 58$ cells per end
gap facts used	$g_k \leq \log 2$ (max. observed 0.405465) and $g_k \geq \log(1 + 1/n_k)$ (max. observed $1/g_k = 1.4985 \cdot 10^5$)
window nesting	$M_{k+1}^{(\text{old})} = M_k^{(\text{new})}$ on all 29 consecutive pairs, an integer identity
non-dyadic gaps	0 of 84 consecutive gap ratios g_k/g_{k+1} is a dyadic power: per-node resolutions have no common refinement
ladder size	52 windows (30 composed + 22 at per-node resolutions, 51 distinct (n, D) pairs), 88 rungs, chains of 2–5 levels, plus 10 bordering-only windows up to $n = 1331$
largest factorisation certificates	$h = 1495$ (cap 1500); finest chain level $h \leq 1400$; base-case sizes $h = 416 \dots 1384$
runtime	8.2 s

Table 1: The verified instance. All entries are outputs of the single script named in Section 7.5.

7.2 The verified instance

Table 1 lists the instance. Table 2 lists what was certified on it. We highlight five entries.

Identities. The nesting of Assumption 2.4 holds to relative $3.0 \cdot 10^{-14}$ for the form and $6.2 \cdot 10^{-16}$ for the pole vector on all 88 rungs; the saturation identity, the dual-residual identity (19) and Galerkin orthogonality hold to $4.7 \cdot 10^{-7}$; the telescope $\sum_l \delta_l = \varepsilon_0 - \varepsilon_L$ to $3.3 \cdot 10^{-7}$; the closed-form piecewise-constant lag formula to $2.2 \cdot 10^{-14}$. The bit-exact part of Lemma 5.2 is bit-exact in the run: the incoming node’s contribution restricted to the old lag range has maximum absolute entry $0.0 \cdot 10^0$, and the trailing block of $Q^{(k+1)}$ agrees with $Q^{(k)}$ to relative $2.9 \cdot 10^{-14}$, which is the roundoff of two different summation orders for the same exact quantity (Lemma 5.3).

The rung. All three certificates of Theorem 4.4 complete on 88 of 88 rungs, and $0 < \hat{c}_l \leq \delta_l$ throughout with retention $\hat{c}_l/\delta_l \in [0.90709, 0.99708]$. The direction control of Remark 3.3 is also verified: the primal form (21), evaluated at the same test vector, returns values *above* δ_l on 88 of 88 rungs, i.e. it is an upper bound and can never serve as a supply.

The lower bound. $\varepsilon_0^{(k)} \geq \sum_l \hat{c}_l^{(k)} > 0$ on all 52 windows. Two normalisations are reported, and the second is the informative one. Against the measured ε_0 the retention is $0.0013 \dots 0.9694$; but an L -level chain cannot supply more than it telescopes, so the relevant ceiling is $1 - \varepsilon_L/\varepsilon_0$, measured at $0.001295 \dots 0.993502$, and against that ceiling the certified fraction is $0.9140 \dots 0.9929$ on every window. The certified margin exceeds the *declared* floating-point bound $c_h u \text{cond}(A_0^{(k)})$ on ε by a factor $31.6 \dots 8.7 \cdot 10^{10}$ at $\text{cond}(A_0) = 6.9 \cdot 10^1 \dots 3.8 \cdot 10^5$. The thinnest window of the ladder is $n = 243$ ($\alpha = 2.7492$, $h_0 = 679$) at 31.6, and it is thin because its retention 0.6054 is pinned to the small ceiling 0.629042 there, not because the arithmetic is marginal.

The base case. At the finest level of each chain, completed Cholesky factorisations of $A^{(L)}$ and of $Q^{(L)} = A^{(L)} - b^{(L)}(b^{(L)})^\top$ both run, at sizes $h = 416 \dots 1384$, with $\lambda_{\min}(Q^{(L)})$ at $4.4 \cdot 10^5 \dots 5.6 \cdot 10^{10}$ times its own floor; and $\text{sign } \lambda_{\min}(Q^{(L)}) = \text{sign } \varepsilon_L$ on 52 of 52 windows, which is (2) verified rather than assumed.

The bordering step. On 62 of 62 steps ($n = 2, \dots, 1331$, $h_{\text{new}} = 30 \dots 1495$) the trailing block is positive definite, the range condition of Lemma 5.5 is vacuous, and the exact $g_c \times g_c$ Schur complement has $\lambda_{\min} \in [1.4839 \cdot 10^{-3}, 1.2528 \cdot 10^{-1}]$, a factor $3.0 \cdot 10^{10} \dots 4.6 \cdot 10^{13}$ above its own floor. The norm-perturbation surrogate of the *same* Schur complement, which divides by $\lambda_{\min}$ of the trailing block, is negative on all 62 steps ($-1.04 \cdot 10^7 \dots -1.29 \cdot 10^1$). Lemma 5.5 never performs that division.

statement	certified / measured on the instance
$P^\top A^{(l+1)} P = A^{(l)}; P^\top b^{(l+1)} = b^{(l)}$	identity; residuals $3.0 \cdot 10^{-14}$ / $6.2 \cdot 10^{-16}$ on 88/88 rungs
$\varepsilon_0 = \varepsilon_L + \sum \delta_l$	identity; residual $3.3 \cdot 10^{-7}$
$\delta_l \geq \hat{c}_l > 0$ (Thm. 4.4)	three completed Choleskys on 88/88 rungs; $\hat{c}_l/\delta_l = 0.90709 \dots 0.99708$
primal form is an upper bound (Rem. 3.3)	above δ_l on 88/88 rungs
$\varepsilon_0 \geq \sum \hat{c}_l > 0$	52/52 windows; 0.9140 $\dots$ 0.9929 of the L -level ceiling; head-room $31.6 \dots 8.7 \cdot 10^{10}$ over the declared floor
base case $Q^{(L)} \succeq 0$	Cholesky at $h = 416 \dots 1384$; head-room $4.4 \cdot 10^5 \dots 5.6 \cdot 10^{10}$
bordering step (Lem. 5.5)	62/62 steps; exact Schur complement $\lambda_{\min} = 1.48 \cdot 10^{-3} \dots 1.25 \cdot 10^{-1}$, head-room $3.0 \cdot 10^{10} \dots 4.6 \cdot 10^{13}$
norm-perturbation surrogate	negative on 62/62 steps
solution-free rung (Prop. 6.1)	$ \rho_l = 0.3888 \dots 14.8018$; norm-only bound returns 0.0 of the rung on 88/88

Table 2: What was certified, and what was measured. Head-room is always a factor over the declared floating-point floor of Section 7.

7.3 Two fitted trends, labelled as fits

The following are least-squares fits with jackknife bands. They are reported for orientation only and are *not* bounds; nothing in Theorem 4.4 or Corollary 5.6 depends on them.

$$\log(\hat{c}_l/\delta_l) = -0.082 - 0.010 \log D + 0.001 \log \alpha, \quad \text{rms } 0.012, \text{ bands } \pm 0.004, \pm 0.005; \quad (32)$$

$$\log\left(\sum_l \hat{c}_l / \sum_l \delta_l\right) = -0.018 - 0.012 \log \alpha, \quad \text{rms } 0.004, \text{ band } \pm 0.002. \quad (33)$$

Equation (33) is fitted on the composed run only, where D is common by construction so no exponent in D is identifiable. Read together, they say that on this ladder the certified fraction of the telescope does not thin out along the run.

7.4 The near-null direction

The measurement that motivates Section 1’s claim about symbol methods is the following, taken on the same family of forms at the two-level stage. Let ℓ be a certified pointwise minorant of the symbol, $\ell \leq \sigma^{(M)}$, so that $T_M(\ell) \preceq T_M(c)$ by Lemma 4.2, and let $E_c = V^\top B^\top T_M(\ell) B V$ be the coarse block it produces. Then $\lambda_{\min}(E_c) < 0$ on every tested row, so the minorant certifies nothing about A_c ; whereas $\lambda_{\min}(A_c)$ itself is positive, measured in $[1.7 \cdot 10^{-5}, 2.9 \cdot 10^{-4}]$, at condition numbers up to $3.1 \cdot 10^7$ — two to five orders of magnitude below the size the minorant would have to reproduce. The near-null eigenvector of A_c is smooth and lies in $\text{ran } V$: its small Rayleigh quotient is produced by cancellation between the Toeplitz entries, not by a small value of the symbol along it, and a pointwise minorant is by construction blind to that. This is exactly the quantity that Theorem 4.4 does not need, since it uses A_c only through $A_c \succeq 0$.

7.5 Reproducibility

All numbers in Tables 1 and 2, and item (iii) and the first band of item (i) of Proposition 6.1, are produced by one self-contained Python script (`grand_assembly_probe.py`, supplied as supplementary material) using only `numpy` and `scipy` (`cho_factor`, `eigvalsh`). It performs no file writes, runs in about 8.2s on a current laptop, and reports 34 internal checks, all passing. Every matrix that is factorised, inverted or diagonalised has $h \leq 1500$; the only larger objects

are matrix-free fast Fourier transforms used to evaluate symbols, at grid size up to 2^{20} . The two remaining sets of numbers — the second band of Proposition 6.1(i), measured on a single level chain rather than on the assembled ladder, and the two-level measurements of Section 7.4 — come from two further scripts of the same family, supplied alongside it; they are reported separately for exactly that reason.

8 Limitations, and what is not claimed

- (L1) **The run is finite, and its length is structurally bounded.** By Definition 5.1 the common resolution is $D = \min_k g_k / (2\nu)$, the window sizes are $M_k \approx u_k / D$, and the cost of a factorisation is cubic in M_k . A longer run therefore forces a smaller D and larger windows, and the run verified here — 30 consecutive nodes, $n = 2, \dots, 79$ — is what fits under a matrix cap of 1500. Nothing is claimed about nodes outside the tested set: of the 34034 comb nodes tabulated up to $n = 400\,000$, exactly 48 were touched at all, and 62 of the 85 admissible bordering steps in that table were certified.
- (L2) **No uniformity in the window index.** This is the main gap, and it is a gap in *uniformity*, not in size. Specifically: hypothesis (A3) of Corollary 5.6 is verified window by window and we have no uniform-in- k lower bound for the $g_c \times g_c$ Schur complement; and we have no uniform lower bound for $\hat{c}_l$. Without both, (30) does not extend to an infinite family, and we make no such claim.
- (L3) **The base case is a certificate, not a theorem.** Hypothesis (A1) is a completed Cholesky factorisation of one explicit matrix of size $h = 416 \dots 1384$ per window. It is verified, not proved.
- (L4) **Convention 4.7 stands.** The certified rung is the carried-solution rung. Proposition 6.1 shows that removing the convention requires a bound on the direction of $\hat{s}_l$, and that norm-based attempts fail by cancellation on the tested ladder; it does not show that no such bound exists.
- (L5) **A second, independent route is deliberately excluded.** The increment δ_l also admits a lower bound through a corner estimate for the oscillation of the Galerkin solution — a one-cell Poincaré step in the spirit of [25] followed by a Cauchy–Schwarz reduction to the corner mass — which requires a sign hypothesis on the increment sequence and a summed Harnack-type ratio bound. Both hold on the windows we measured and neither is proved uniformly; their classical address is the corner regularity of inverses of Toeplitz forms with singular symbols [30, 14, 28] for the ratio, and the discrete maximum principle for monotone matrices [24, 29] for the sign. Since that route is logically independent of Theorem 4.4 and Corollary 5.6, which do not use it, we omit it here and defer it to a separate note whose subject is a discrete Harnack inequality for the increments of the inverse.
- (L6) **Fits are fits.** Equations (32) and (33), and every exponent quoted for orientation, are least-squares fits with jackknife bands, not bounds.
- (L7) **The example class is an example.** Theorem 4.4 and Corollary 5.6 are symbol-agnostic; the class of Definition 2.6 enters only through the verification, and the arithmetic in it enters only as the source of the comb positions and through the two classical gap inequalities named in Section 2.4(iii). No asymptotic property of the symbol family and no arithmetic statement is used or claimed.

Acknowledgements

The construction grew out of a long sequence of failed attempts, and the two that were most instructive are recorded above: the primal form of a rung (Remark 3.3, an upper bound where a lower bound was wanted) and the norm-perturbation surrogate of the bordering step (Section 5, negative on every step). Both are reported as measurements because both are, in retrospect, the reason the working argument has the shape it has.

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